Test record: how far does one wheel rotation go?

VisuBot · lesson “Your own methods” · overloading mbot.forward(value, "rotations")

Name _____ Class _____ Date _____ Robot no. _____

What is this about? VisuBot assumes a wheel diameter of **6.5 cm**. One wheel rotation is then π · 6.5 cm ≈ **20.4 cm**. Is that true for *your* mBot2? You test the software like a professional: you compare the **expected** result (what the program promises) with the **actual** result (what you measure).

Material: mBot2, ruler or tape measure, tape for the start line, smooth floor.

A · Measure the wheel

Wheel diameter (with tyre) d = _____ cm

Circumference C = π · d = _____ cm

Difference from 20.4 cm: _____ cm

B · Driving test

Mark the start line at the wheel axle. Run the program and measure how far the axle has moved.

Test case	Program	Expected	Actual (measured)	Difference
1	`mbot.forward(5, "rotations");`	102.1 cm		
2	`mbot.forward(5, "rotations");`	102.1 cm		
3	`mbot.forward(5, "rotations");`	102.1 cm		
4	`mbot.forward(100, "cm");`	100.0 cm		
5	`mbot.backward(2, "rotations");`	40.8 cm		

C · Evaluation

Average of test cases 1–3: _____ cm

Difference in percent = (actual − expected) : expected · 100 = _____ %

Result: ☐ passed (difference below 3 %) ☐ failed

Possible causes of a difference (tick):

☐ slippery floor or carpet
☐ weak battery
☐ dirty or worn tyres
☐ crooked start / inaccurate measuring
☐ different wheel diameter

My feedback for the teacher (what did you notice?):

This is how software systems are tested. Every **test case** has an input (the program), an **expected result** and an **actual result**. You test **several times**, because one try can be chance. Everything goes into a **test record** so others can repeat the test. If the actual result differs, you write a **bug report** – which is exactly what this sheet is.